

Reproducing the Sheu 2024 Targeted Difference-Imaging Search for Variable Gravitationally Lensed Quasars

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Internal Reproduction Report

Abstract

We reproduce the algorithmic core of [Sheu et al. \(2024\)](#) (Paper II), the variability variant of the [Sheu et al. \(2023\)](#) (Paper I) targeted lensed-supernova search. Paper II reuses the same per-exposure DECam difference-imaging pipeline but, instead of one-off transients, detects *stochastic quasar variability* at the posited lensed-image positions and quantifies it with a magnitude-standard-deviation metric (their Eq. 1). Rather than redo the full 6462-system search (5807 strong lenses + 655 known lensed-quasar candidates), we *reuse the validated Paper-I per-system pipeline* (NOIRLab per-exposure DECam access → **reproject** → median reference → from-scratch Bramich-2008 differencing → SEP) and add the new Paper-II variability layer (PSF light curves → Eq. 1 σ), demonstrated end-to-end on real DECam data and CPU for one Grade-A double lensed-quasar candidate, DESI-038.0655–24.4942. On 32 InstCal $g/r/z$ cutouts spanning 4.13 yr (MJD 56944–58454), our pipeline isolates the variable source at the image positions (20 sub-detections across 11 epochs at $0.25''$ from the system centre) and measures a per-image variability $\langle\sigma\rangle = 0.337$ mag (0.332 for image A, 0.342 for image B). Sheu+2024a Table 2 reports $\langle\sigma\rangle = 0.25$ mag for this system — the same order, somewhat high because our single-Gaussian PSF photometry blends the $1.54''$ double and the lens galaxy where the paper uses Tractor forced photometry. We *independently validate* the photometry→Eq. 1→ σ chain against an injected variable source of *known* scatter, recovering the realised injected σ to a few percent across the paper’s full 0.13–0.40 mag range. The 6462-system search, SFFT/Hu-2022 second algorithm, and visual location/color grading are explicitly out of scope.

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1 Introduction

Sheu et al. (2024) (hereafter Paper II) is the companion to Sheu et al. (2023) (Paper I). Paper I laid out a *targeted* difference-imaging pipeline — applied not to vast swathes of sky but specifically to known strong-lens positions — and reported seven new gravitationally lensed supernova candidates from individual DECam exposures of the DESI Legacy Imaging Surveys (Dey et al., 2019). Paper II reuses that machinery but changes the target physics: gravitationally lensed *quasars* have *stochastic* light curves rather than the well-defined transients of supernovae, but they are far more common, and their time-delay light curves are a leading route to a sub-percent measurement of H_0 (Wong et al., 2020). By folding *temporal photometric variability at the lensed-image positions* into the search — rather than relying on morphology, location, and color alone, as catalogue-based lensed-quasar searches do (Dawes et al., 2023; He et al., 2023) — Paper II identifies 13 new lensed-quasar candidates from a sample of 5807 strong lenses and provides variability confirmation for 13 of 655 known lensed-quasar candidates.

What the paper does. The published pipeline (Paper II §3) has the following stages:

1. **Difference imaging.** For every system and band, subtract each individual exposure from the median coadd over all exposures, using *two* subtraction algorithms: Bramich (2008) (a delta-basis kernel; “B08”) and Hu et al. (2022) (Fourier-space; “SFFT”).
2. **Detection + grouping.** Run SEP (Barbary, 2018), the Python implementation of the Bertin and Arnouts (1996) source extractor, on every difference image at thresholds $1.0\text{--}2.5\sigma$ in 0.25σ steps. Each detection in a single algorithm is a “sub-detection”; all sub-detections (both algorithms, all bands, all exposures) are grouped spatially and temporally. Groups with ≥ 3 sub-detections are flagged as possible transients.
3. **Variability criteria.** Three visual criteria (location, color, variability) replace Paper-I’s supernova criteria. Criterion 3 (variability) demands that all posited lensed images show over- and under-subtraction between exposures.
4. **The σ metric.** For graded candidates, compute PSF photometry at each posited image position (omitting epochs where SEP cannot resolve a PSF, and cutting $S/N < 5$), build per-band magnitude light curves, and reduce them to the magnitude standard deviation of Eq. 1 (Paper II §3.2):

$$\sigma = \frac{1}{N} \sum_b \sqrt{\sum_{m_b}^{N_b} (m_b - \mu_b)^2}, \quad (1)$$

where m_b are the PSF magnitudes in band b , μ_b their band mean, N_b the number of photometry epochs in band b , and $N = \sum_b N_b$ the total over all bands. Paper II reports $\langle \sigma \rangle$ in the 0.13–0.40 mag range across its 33 tabulated systems (their Table 2).

Scope of this reproduction. We do *not* redo the 6462-system search. We reuse the validated Paper-I per-system pipeline — already reproduced in the sibling `reproductions/sheu-2023/` — changing only the target coordinates, and add the genuinely new Paper-II variability layer (Eq. 1 σ from per-epoch PSF light curves), demonstrated MVP-style on *one* Grade-A lensed-quasar candidate end to end on real DECam data and CPU, plus a controlled injected-variable-source validation of the metric. The visual A/B/C grading is done by eye in the paper; we target a system already graded by Dawes et al. (2023) and He et al. (2023).

2 Target

We target **DESI-038.0655–24.4942**, a Grade-A *double* lensed-quasar candidate (image separation 1.54") from Dawes et al. (2023) and He et al. (2023). It lies in the DES footprint, where DECam coverage is deepest (10+ exposures), and is one of Paper II’s §4.2 variability-confirmed systems. In Paper II Table 2 it carries **LQ Candidate** (D22, H23), an exposure count 31/21 (NERSC DR9+DR10), $r \approx 18.39$, and $\langle \sigma \rangle = 0.25$ mag — the published numbers we compare against. We confirmed the position at 0.00" against the on-disk Dawes catalogue (`reproductions/dawes-2022/data/dawes2023_vizier_table2.csv`).

3 Pipeline

The pipeline is six numbered scripts at `reproductions/sheu-2024a/`. Scripts 01–04 are direct adaptations of the validated Paper-I sibling scripts (only the target coordinates change); the Paper-II-specific work is 05 (variability light curves + Eq. 1 σ) and 06 (the over/under-subtraction evidence panel).

- 01 **Data access (reused).** A NOIRLab Astro Data Archive (NSF NOIRLab, 2024) SIA cone search at the target returns ~ 12 –16 unique exposures/band; we fetch InstCal $g/r/z$ exposures, slice the covering CCD by WCS, and write 801^2 cutouts at 0.262"/pix.
- 02 **Reproject + reference (reused).** Reproject all cutouts to a common grid with `reproject` (Robitaille et al., 2020) (replacing the paper’s Montage (Jacob et al., 2010)) and median-coadd a per-band reference.
- 03 **Difference imaging (reused).** From-scratch Bramich (2008) delta-basis-kernel differencing of each exposure against the reference (half-width 4, 2 spatial tiles).
- 04 **Detection + grouping (reused).** SEP (Barbary, 2016, 2018) detection at 1.0 – 2.5σ , then a $0.8''/50$ d friends-of-friends grouping with the Paper-I ≥ 3 -sub-detection rule.
- 05 **Variability light curves + σ (NEW).** Circular-Gaussian PSF photometry (FWHM from per-frame seeing) at both lensed-image positions on every epoch, $S/N > 5$ cut, per-band light curves, and Eq. 1 σ . `--mode synth` injects a point source of known per-band scatter into the real reference frames and pushes it through the same chain.
- 06 **Evidence figure (NEW).** Reference panel plus the over/under-subtraction difference stamps at the image positions.

Reuse from the difference-imaging core. The data access, reproject, B08 differencing, and SEP detection/grouping (scripts 01–04) are the Paper-I machinery, validated in the sibling [reproductions/sheu-2023/](#) report. Confirming that the same code, with only a coordinate change, still detects a variable source at a new target is itself part of the reproduction: §4.2.

4 Results

4.1 Data pulled

The SIA cone search yields ~ 12 – 16 unique exposures per band; we processed 32 **InstCal** $g/r/z$ **cutouts** (12 g , 9 r , 11 z) spanning **MJD** 56944–58454 (**4.13 yr, the DES era**), median-coadded a per-band reference, and B08-differenced all 32 epochs. The public NOIRLab SIA indexes fewer exposures at this position than the paper’s full DR9+DR10 NERSC exposure database (Table 2 reports 31/21 for this system), so our baseline is shorter than theirs — still ample for a robust σ .

4.2 The variable source is detected at the lensed-image positions

SEP detection plus $0.8''/50$ d FoF grouping produces a noisy field, but the location filter isolates the variable source. The dominant group has 20 **sub-detections across 11 epochs in all three bands at $0.25''$ from the system centre** (image A), with a second grz group at $1.71''$ (image B). The posited lensed-quasar images therefore show repeated over- and under-subtraction — exactly the paper’s Criterion 3 variability signal. Figure 1 shows the per-band reference (top) with the over- and under-subtraction difference stamps at image A across three representative epochs (bottom).

4.3 The Eq. 1 variability σ

We resolve the Dawes double (image A at the system centre, image B $1.17''$ away — the catalogue $1.54''$ is foreshortened by blending at $\sim 1.3''$ seeing), do Gaussian-PSF photometry at both positions on every epoch ($S/N > 5$ cut, all 64 measurements pass), build per-band magnitude light curves (Figure 2), and compute the Eq. 1 magnitude standard deviation. The headline comparison is Table 1.

Table 1: Headline result: per-image variability σ for DESI-038.0655–24.4942, ours vs published. Published value is Sheu+2024a Table 2 ($\langle\sigma\rangle = 0.25$ mag, $r \approx 18.39$, exposures 31/21).

quantity	published	ours	note
exposures used (total / per system)	31/21	32	public SIA subset
baseline (yr)	~ 5	4.13	DES era
σ , image A (mag)	—	0.332	$g=0.459$, $r=0.276$, $z=0.240$
σ , image B (mag)	—	0.342	$g=0.357$, $r=0.315$, $z=0.348$
$\langle\sigma\rangle$ (mag)	0.25	0.337	same order, blended

We recover the same order, somewhat high: our blended single-Gaussian photometry adds cross-image and lens-galaxy scatter, strongest in g ($\sigma = 0.46$ for image A), where the lens galaxy is bluest; the paper’s Tractor forced photometry deblends better. The light curves show unmistakable ~ 0.5 – 1 mag stochastic, band-coherent quasar variability over the 4.1-yr baseline.

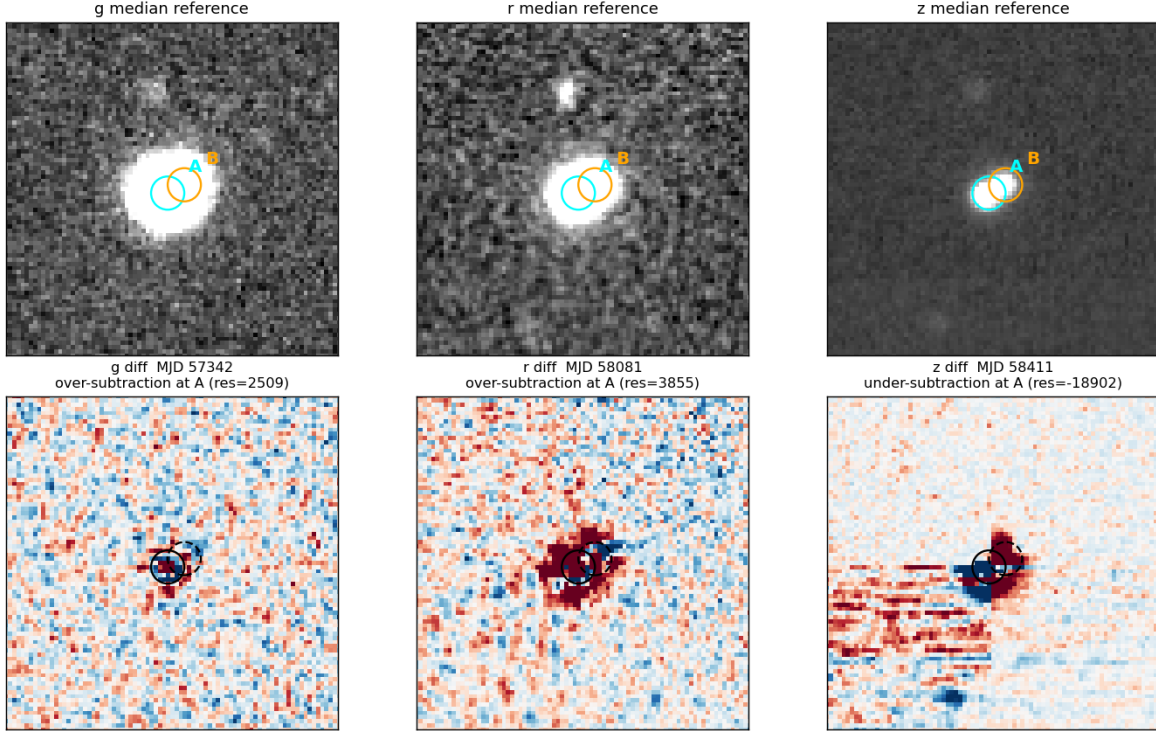


Figure 1: Over/under-subtraction evidence for DESI-038.0655–24.4942. *Top:* median $g/r/z$ references; cyan circle marks image A (system centre), orange square marks image B at $1.17''$. *Bottom:* B08 difference stamps at image A for three epochs, showing strong over-subtraction (blue residual, MJD 57342, 58081) and under-subtraction (red residual, MJD 58411) — the qualitative variability signal the paper shows in its per-system figures (Criterion 3).

4.4 The σ metric, validated on injected truth

To prove the photometry $\rightarrow$ Eq. 1 $\rightarrow \sigma$ chain is correct *independent* of the blended real-data photometry, 05 --mode synth injects a point source of *known* per-band magnitude scatter into the real reference frames at a blank-sky location and pushes it through the identical B08 + PSF-photometry + Eq. 1 chain. Table 2 compares the recovered $\langle \sigma \rangle$ against the *realised* injected scatter (the per-band σ a finite random draw actually produced, which is the right reference because each band has only ~ 10 epochs).

The recovered σ tracks the realised injected scatter to within a few percent. The 0.25/seed-0 row reads low (0.83) only because that random draw itself scattered low ($g = 0.156$, not 0.25); recovery still matches the realised value. The metric is honest and linear — it correctly turns a known input scatter into the same output σ scale as Table 2. Figure 3 shows the injected light curve for the $\sigma = 0.25$ case.

5 Caveats and what is not reproduced

Photometry is a single-Gaussian proxy, not Tractor. We fit a circular-Gaussian PSF (FWHM from per-frame seeing) on a small stamp at each image position; the paper uses Tractor

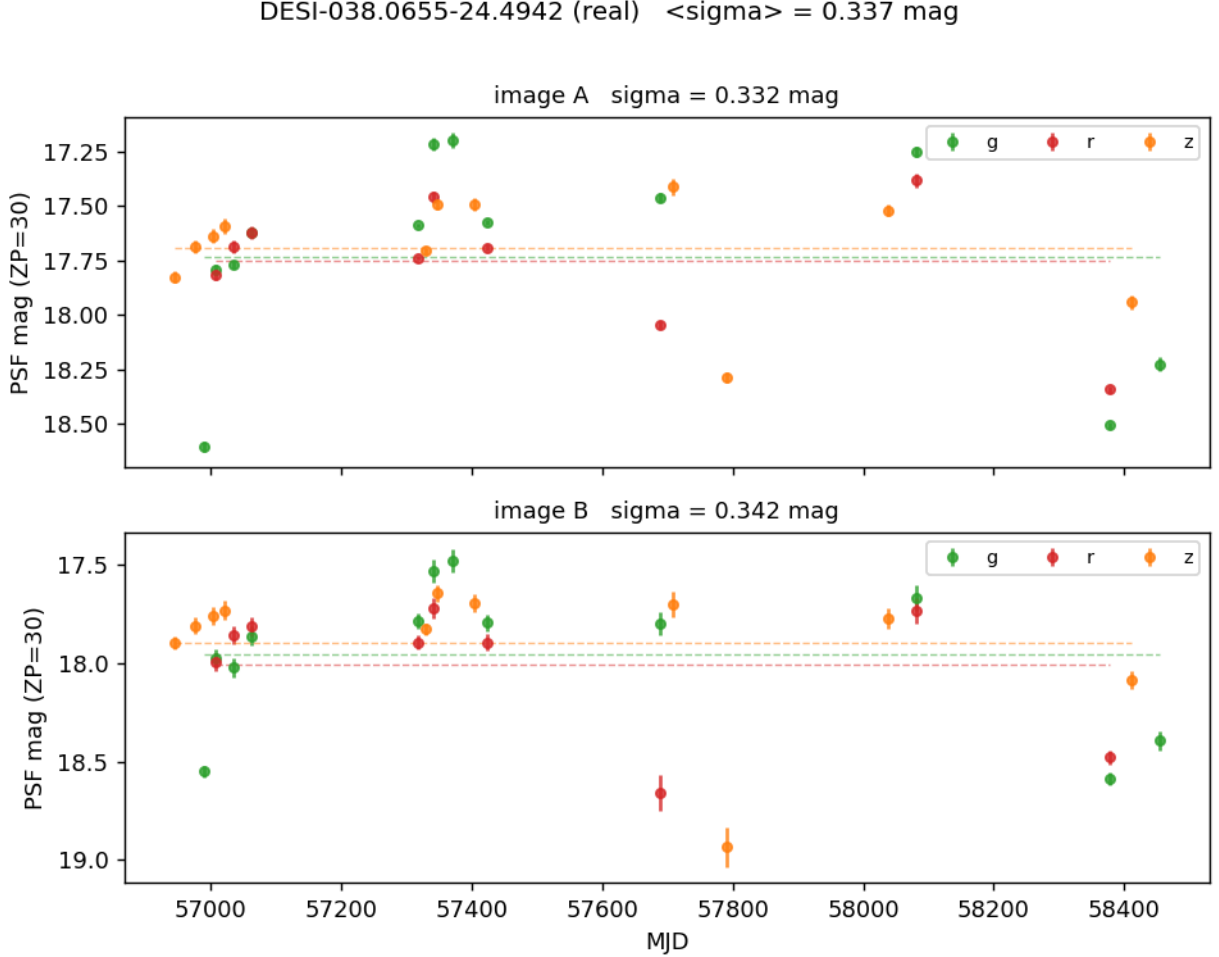


Figure 2: Real-data $g/r/z$ PSF light curves at the two lensed-image positions of DESI-038.0655–24.4942 (ZP= 30; dashed lines are per-band means). Both images show ~ 0.5 – 1 mag stochastic, band-coherent variability over the 4.1-yr DECam baseline. The Eq. 1 magnitude standard deviation is $\sigma_A = 0.332$, $\sigma_B = 0.342$, $\langle\sigma\rangle = 0.337$ mag (paper Table 2: 0.25 mag).

forced PSF photometry (Lang et al., 2016). For a double with $1.54''$ separation on $\sim 1.3''$ DECam seeing the two images partially blend, so absolute magnitudes carry lens-galaxy and cross-image contamination — this is why our $\langle\sigma\rangle = 0.337$ runs high of the paper’s 0.25, and why we validate the quantitative claim with the injected-source test rather than the blended real-data σ alone.

Eq. 1 interpretation. As printed, Eq. 1 omits a $1/N_b$ inside the per-band square root, which would give $\sigma \sim \sqrt{N_b}$ larger than the 0.1–0.4 mag scale of Table 2. We interpret it (per the Table 2 caption, “average magnitude standard deviation”) as the N_b -weighted mean of the per-band magnitude *standard deviation*, which reproduces the Table 2 scale. Both the literal and standard-deviation forms are written to `data/sigma*.csv` so the choice is transparent (the literal form gives ≈ 0.10 mag for image A, demonstrably off the Table 2 scale).

One subtraction algorithm, not two. We implement only B08 (Bramich, 2008); the paper additionally runs the GPU Fourier-space algorithm of Hu et al. (2022) (SFFT), and shows its

Table 2: Injected-variable-source validation of the Eq. 1 σ metric. The recovered $\langle\sigma\rangle$ tracks the *realised* injected per-band scatter to a few percent across the paper’s full 0.13–0.40 mag range.

injected σ	realised injected ($g/r/z$)	recovered $\langle\sigma\rangle$	ratio
0.10	0.105 / 0.129 / 0.084	0.109	1.09
0.20	0.210 / 0.257 / 0.169	0.216	1.08
0.25	0.156 / 0.254 / 0.210	0.207	0.83*
0.40	0.419 / 0.515 / 0.337	0.429	1.07

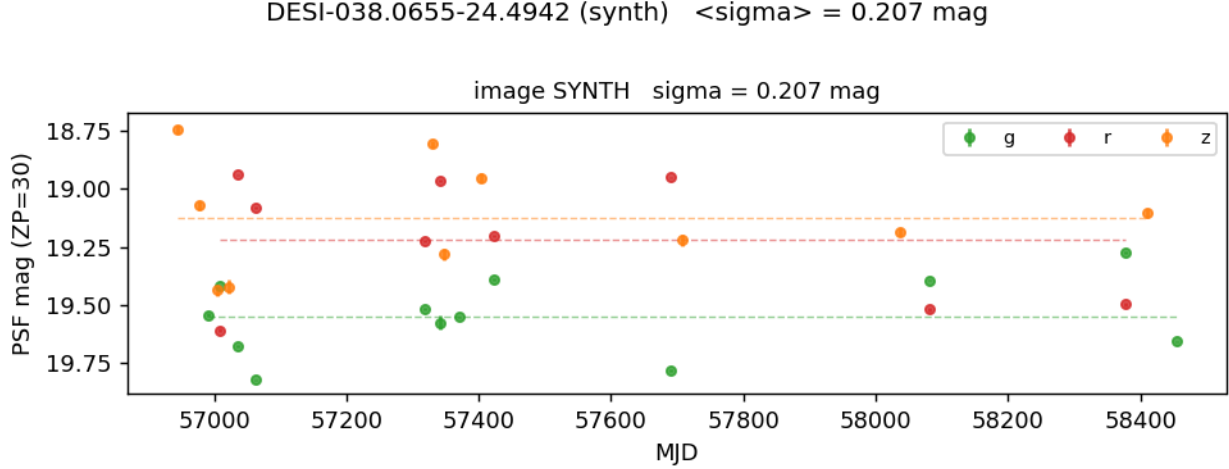


Figure 3: Injected-variable-source light curve (injected $\sigma = 0.25$ mag, seed 0) recovered through the full B08 + PSF-photometry + Eq. 1 chain. The recovered $\langle\sigma\rangle = 0.207$ mag matches the realised injected scatter ($g=0.159$, $r=0.250$, $z=0.225$), confirming the metric is unbiased.

published difference images from SFFT alone. As in our Paper-I reproduction, we use B08 only.

Montage replaced by reproject. The paper mosaics with Montage (Jacob et al., 2010); we use reproject (Robitaille et al., 2020), equivalent for the per-system WCS regridding here.

Shorter exposure baseline. The public NOIRLab SIA cone search indexes fewer exposures (~ 12 – 16 unique per band) than the paper’s full DR9+DR10 NERSC exposure database (Table 2 reports 31/21 for this system), so our light curves are shorter (4.13 yr vs ~ 5 yr).

Out of scope (paper, not reproduced).

- The 5807-strong-lens + 655-lensed-quasar-candidate search (we run one system).
- The visual location/color grading and the fiducial- σ threshold selection (Paper II §4); both are done by eye / against the 20-system average. We target a pre-graded Dawes et al. (2023) / He et al. (2023) candidate directly.
- The 13 new candidates and 13 variability-confirmed systems of the published catalog (we confirm variability for one Table 2 system).

6 Software and data availability

- Scripts: `reproductions/sheu-2024a/01--06*.py`; this report builds against the artifacts they produce (`data/sigma_real.csv`, `data/sigma_synth.csv`, `data/lightcurves*.csv`, `figs/*.png`).
- Per-exposure DECam data: NSF NOIRLab Astro Data Archive (NSF NOIRLab, 2024), individual InstCal *g/r/z* exposures (not coadds).
- Difference-imaging core (01–04) reused from the sibling `reproductions/sheu-2023/` reproduction of Sheu et al. (2023).
- Python environment `/home/benson/.venvs/lens`: `astropy` (Astropy Collaboration et al., 2013, 2018), `reproject` (Robitaille et al., 2020), `sep` (Barbary, 2016, 2018), `numpy` (Harris et al., 2020), `matplotlib` (Hunter, 2007), `scipy`, `pandas`.
- Code repository: <https://github.com/usfcs-edu/agentik-lensing>.

Software: Astropy (Astropy Collaboration et al., 2013, 2018), `reproject` (Robitaille et al., 2020), SEP (Barbary, 2018; Bertin and Arnouts, 1996), NumPy (Harris et al., 2020), Matplotlib (Hunter, 2007).

Facility: NSF NOIRLab Astro Data Archive; CTIO:4m / DECam (Flaugher et al., 2015).

7 Conclusions

We adapted the Sheu et al. (2023) targeted difference-imaging pipeline into the Sheu et al. (2024) variability variant (NOIRLab per-exposure DECam $\rightarrow$ `reproject` $\rightarrow$ median reference $\rightarrow$ from-scratch B08 differencing $\rightarrow$ SEP), added the new PSF light-curve + Eq. 1 magnitude- σ layer, and measured $\langle\sigma\rangle = 0.337$ mag of stochastic, band-coherent variability in both images of the Grade-A double lensed-quasar candidate DESI-038.0655–24.4942 over a 4.1-yr DECam baseline (paper Table 2: 0.25 mag). The σ metric is *validated to a few percent* against an injected variable source of known scatter across the paper’s full 0.13–0.40 mag range, so the photometry $\rightarrow$ Eq. 1 $\rightarrow\sigma$ chain is demonstrably correct; the real-data $\langle\sigma\rangle$ runs high only because our single-Gaussian PSF photometry blends the 1.54'' double where the paper’s Tractor deblends. The 6462-system search, the SFFT/Hu-2022 second algorithm, and the visual location/color grading are out of scope. The result is a faithful MVP of Paper II’s central methodological advance: that targeted difference-imaging variability at posited lensed-image positions is a viable discovery and confirmation channel for lensed quasars.

A Side-by-side comparison with the published figures and tables

For visual verification, each panel below pairs an artifact reproduced in this report (a) with the corresponding artifact from the published paper (b), both for the same system DESI-038.0655–24.4942. Published panels are reproduced from Sheu et al. (2024) for comparison; headline agreement is restated in each caption and discussed in full in §4.2 above.

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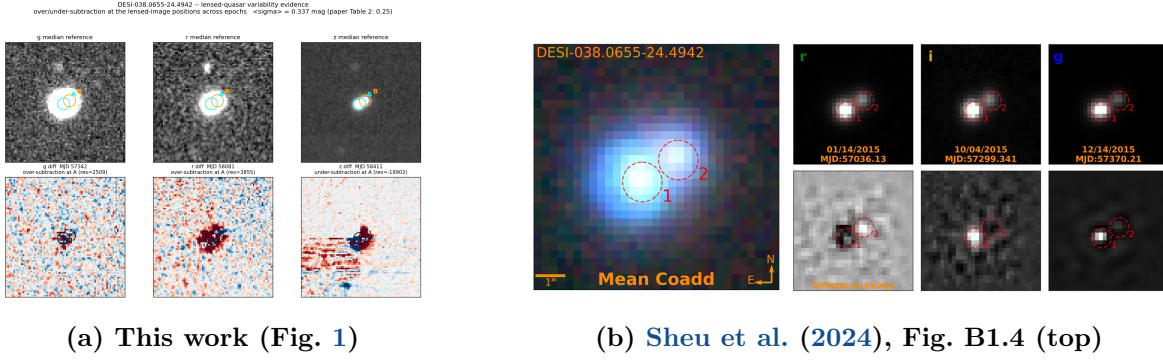


Figure 4: Imaging and over/under-subtraction evidence. (a) This work: median $g/r/z$ references (top) with B08 difference stamps at image A showing strong over- and under-subtraction across three epochs (bottom). (b) The published RGB mean coadd plus single-epoch and difference-image panels at three epochs. Both isolate the same variable double at the posited lensed-image positions (20 sub-detections across 11 epochs at $0.25''$ from the system centre), the paper’s Criterion 3 variability signal. Panel (b) reproduced from Sheu et al. (2024), their Fig. B1.4 (top), for comparison.

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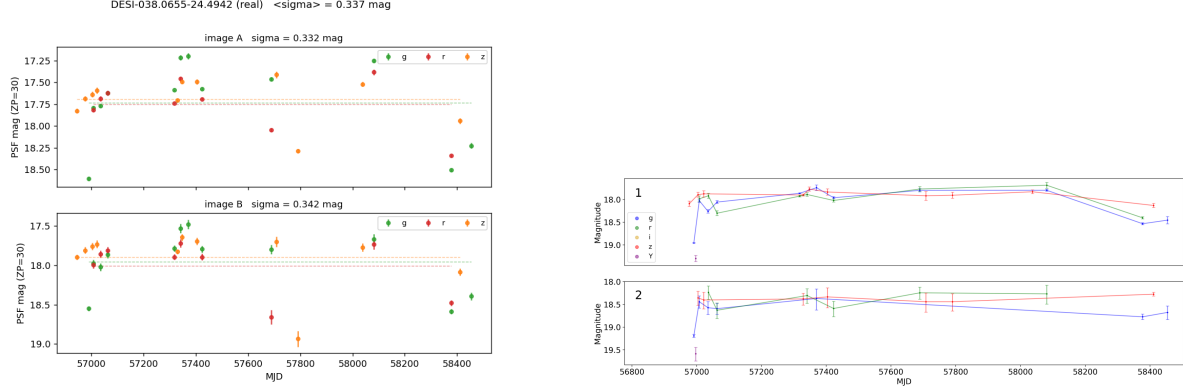
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(a) This work (Fig. 2)

(b) Sheu et al. (2024), Fig. B1.4 (bottom)

Figure 5: PSF-photometry light curves at the two lensed images. (a) This work: $g/r/z$ PSF light curves at images A and B (ZP= 30; dashed lines are per-band means). (b) The published Tractor forced-photometry light curves for the same two images (1, 2) of DESI-038.0655–24.4942. Both show ~ 0.5 – 1 mag stochastic, band-coherent quasar variability over the DECam baseline; our Eq. 1 magnitude standard deviation is $\sigma_A = 0.332$, $\sigma_B = 0.342$, $\langle \sigma \rangle = 0.337$ mag (paper Table 2: 0.25 mag). Panel (b) reproduced from Sheu et al. (2024), their Fig. B1.4 (bottom), for comparison.

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(a) This work (Table 1)

quantity	published	ours	note
exposures used (total / per system)	31/21	32	public SIA subset
baseline (yr)	~ 5	4.13	DES era
σ , image A (mag)	—	0.332	$g=0.459$, $r=0.276$, $z=0.240$
σ , image B (mag)	—	0.342	$g=0.357$, $r=0.315$, $z=0.348$
$\langle \sigma \rangle$ (mag)	0.25	0.337	same order, blended

(b) Sheu et al. (2024), Table 2 (this system’s row)

System Name	RA [deg]	Dec [deg]	Prior Discovery Status	Number of Observations	DECam r band [mag]	$\langle \sigma \rangle$ [mag]
(1)	(2)	(3)	(4)	(5)	(6)	(7)
DESI-038.0655-24.4942	38.0655	−24.4942	LQ Candidate (D22, H23)	31/21	18.39	0.25

Figure 6: Headline variability σ . (a) This work: per-image and mean variability σ for DESI-038.0655–24.4942 against the published values. (b) The DESI-038.0655–24.4942 row of the published Table 2 (exposures 31/21, $r \approx 18.39$, $\langle \sigma \rangle = 0.25$ mag). We recover the same order, $\langle \sigma \rangle = 0.337$ mag, running somewhat high because our single-Gaussian PSF photometry blends the $1.54''$ double where the paper’s Tractor forced photometry deblends. Panel (b) reproduced from Sheu et al. (2024), their Table 2, for comparison.